

The tikz package

This is a general purpose graphics package.¹ To load it for this document, I used:

```
\usepackage{tikz}
\usetikzlibrary{matrix,arrows,decorations.pathmorphing}
```

With `tikz` you can probably produce any diagram you need with any precision you desire, but the code can be complex. You may not wish to use it for a simple diagram:

$$\begin{array}{ccc} A & \xrightarrow{a} & B \\ \downarrow b & & \downarrow c \\ C & \xrightarrow{d} & D \end{array}$$

$$\begin{array}{ccc} A & \xrightarrow{a} & B \\ \downarrow b & & \downarrow c \\ C & \xrightarrow{d} & D \end{array}$$

```
\begin{tikzpicture}
\matrix(m)[matrix of math nodes,
row sep=2.6em, column sep=2.8em,
text height=1.5ex, text depth=0.25ex]
{A&B\\
C&D\\};
\path[->,font=\scriptsize,>=angle 90]
(m-1-1) edge node[auto] {$a$} (m-1-2)
edge node[auto] {$b$} (m-2-1)
(m-1-2) edge node[auto] {$c$} (m-2-2)
(m-2-1) edge node[auto] {$d$} (m-2-2);
\end{tikzpicture}
```

```
\begin{CD}
A @>a>> B \\
@VVbV @VVcV \\
C @>d>> D
\end{CD}
(amscd)
```

This code sets up a matrix named `m` with some options, and then places A , B , C , and D at the four positions of a 2×2 matrix. The next line specifies normal arrows with labels in `scriptsize` and a nondefault arrow head, and the following line specifies an arrow from the (1,1) position of the matrix `m` to the (1,2) position with a label a in the default position.

However, `tikz` handles large objects and tall labels better than `amscd`:

$$\begin{array}{ccc} A \times A \times A \times A \times A \times A & \xrightarrow{a} & B \\ \downarrow b & & \downarrow c \\ C & \xrightarrow{d} & D \end{array}$$

$$\begin{array}{ccc} A & \xrightarrow{\quad} & B \\ \downarrow & & \downarrow \\ C & \xrightarrow{\overline{A^{A^A}}} & D \end{array}$$

To my eyes, the arrow heads are too small. This can be fixed by adding `>=angle 90`, as an option to the `path` or to the whole picture:

```
————— \path[->](1,1) edge (2,1);
————— \path[->,>=angle 90](1,1) edge (2,1);
```

This is part of: Guide to Commutative Diagrams, www.jmilne.org/not/CDGuide.html
 March 7, 2011

¹Felix Lenders has posted a tutorial “Commutative Diagrams using TikZ” at <http://www.felixl.de/commu.pdf>.

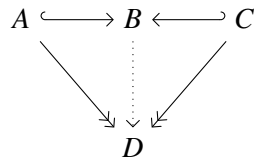
Here is the code for some arrows.

```

—————> \path[->](1,1) edge (2,1);
|—————> \path[|->](1,1) edge (2,1);
————— \path[-](1,1) edge (2,1);
<———— \path[right hook->](1,1) edge (2,1);
—————>> \path[->>](1,1) edge (2,1);
.....> \path[dotted,->](1,1) edge (2,1);
-----> \path[dashed,->](1,1) edge (2,1);
●————> \path[*->](1,1) edge (2,1);
~~~~~> http://tex.stackexchange.com/questions/12678/

```

The next example illustrates the use of the different arrows in a commutative diagram

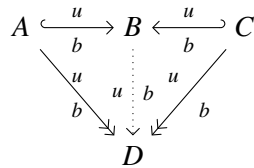


```

\begin{tikzpicture}[>=angle 90]
\matrix(a)[matrix of math nodes,
row sep=3em, column sep=2.5em,
text height=1.5ex, text depth=0.25ex]
{A&B&C\\
&D\\};
\path[right hook->](a-1-1) edge (a-1-2);
\path[->>](a-1-1) edge (a-2-2);
\path[dotted,->](a-1-2) edge (a-2-2);
\path[left hook->](a-1-3) edge (a-1-2);
\path[->>](a-1-3) edge (a-2-2);
\end{tikzpicture}

```

Now an example with labels on the arrows:



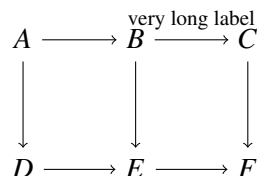
```

\begin{tikzpicture}[>=angle 90]
\matrix(a)[matrix of math nodes, row sep=3em, column sep=2.5em,
text height=1.5ex, text depth=0.25ex]
{A&B&C\\
&D\\};
\path[right hook->,font=\scriptsize]
(a-1-1) edge node[above]{$u$} node[below]{$b$} (a-1-2);
\path[->>,font=\scriptsize]
(a-1-1) edge node[above]{$u$} node[below]{$b$} (a-2-2)
(a-1-3) edge node[above left]{$u$} node[below right]{$b$} (a-2-2);
\path[dotted,->,font=\scriptsize]
(a-1-2) edge node[left]{$u$} node[right]{$b$} (a-2-2);
\path[left hook->,font=\scriptsize]

```

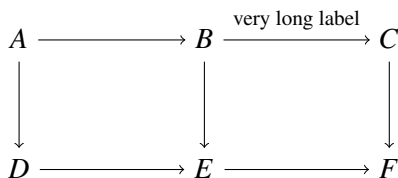
```
(a-1-3) edge node[above]{$u$} node[below]{$b$} (a-1-2);
\end{tikzpicture}
```

Long labels may cause a problem:



```
\begin{tikzpicture}
\matrix(m)[matrix of math nodes,
row sep=3em, column sep=2.5em,
text height=1.5ex, text depth=0.25ex]
{A&B&C\\
D&E&F\\};
\path[->,font=\scriptsize]
(m-1-1) edge (m-1-2)
edge (m-2-1)
(m-1-2) edge node[auto] {very long label} (m-1-3)
edge (m-2-2)
(m-1-3) edge (m-2-3)
(m-2-1) edge (m-2-2)
(m-2-2) edge (m-2-3);
\end{tikzpicture}
```

However, this can be fixed by setting `column sep=5.0em`.



Nor does it have a problem with objects of different heights.

$$\hat{A} \longrightarrow \prod_{n \in \mathbb{Z}} A_n \longrightarrow \prod_{n \in \mathbb{Z}} A_n.$$

But that is because of the options `text height=1.5ex`, `text depth=0.25ex`. When you omit them, you get:

$$\hat{A} \longrightarrow \prod_{n \in \mathbb{Z}} A_n \longrightarrow \prod_{n \in \mathbb{Z}} A_n.$$

Curving arrows is easy.

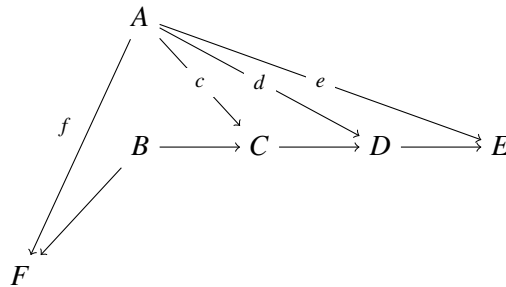


```

\begin{tikzpicture}
\matrix(m)[matrix of math nodes,
row sep=3em, column sep=2.8em,
text height=1.5ex, text depth=0.25ex]
{A&B\\};
\path[->]
(m-1-1) edge [bend left] (m-1-2)
edge [bend left=40] (m-1-2)
edge [bend left=60] (m-1-2)
edge [bend left=80] (m-1-2)
edge [bend right] (m-1-2);
\end{tikzpicture}

```

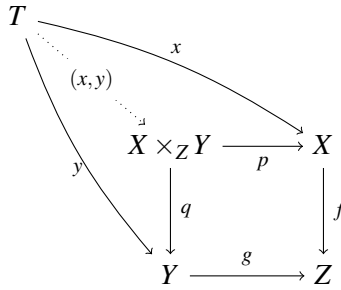
Two more examples:



```

\begin{tikzpicture}[descr/.style={fill=white}]
\matrix(m)[matrix of math nodes, row sep=3em, column sep=2.8em,
text height=1.5ex, text depth=0.25ex]
{&A\\&B&C&D&E\\F\\};
\path[->,font=\scriptsize]
(m-1-2) edge node[above left] {$f$} (m-3-1)
edge node[descr] {$c$} (m-2-3)
edge node[descr] {$d$} (m-2-4)
edge node[descr] {$e$} (m-2-5);
\path[->]
(m-2-2) edge (m-3-1)
edge (m-2-3);
\path[->]
(m-2-3) edge (m-2-4);
\path[->]
(m-2-4) edge (m-2-5);
\end{tikzpicture}
\end{tikzpicture}

```



```
\[
\begin{tikzpicture}[descr/.style={fill=white}]
\matrix(m)[matrix of math nodes, row sep=3em, column sep=2.8em,
text height=1.5ex, text depth=0.25ex]
{T\\&X\times_Z Y&X\\&Y&Z\\};
\path[->,font=\scriptsize]
(m-1-1) edge [bend left=10] node[above] {$x$} (m-2-3)
(m-1-1) edge [bend right=10] node[below] {$y$} (m-3-2);
\path[->,dotted,font=\scriptsize]
(m-1-1) edge node[descr] {$$(x,y)$`} (m-2-2);
\path[->,font=\scriptsize]
(m-2-2) edge node[below] {$p$} (m-2-3)
(m-2-2) edge node[right] {$q$} (m-3-2);
\path[->,font=\scriptsize]
(m-2-3) edge node[right] {$f$} (m-3-3);
\path[->,font=\scriptsize]
(m-3-2) edge node[above] {$g$} (m-3-3);
\end{tikzpicture}
\]
```

When you number a displayed commutative diagram

```
\begin{equation}
\begin{tikzpicture}
.....
\end{tikzpicture}
\end{equation}
```

$$\begin{array}{ccccc}
 A & \xrightarrow{a} & B & \xrightarrow{b} & C \\
 \downarrow c & & \downarrow d & & \downarrow e \\
 D & \xrightarrow{f} & E & \xrightarrow{g} & F
 \end{array}$$

(1)

the number appears below the level of the diagram. To centre the number, use:

```

\begin{equation}
\begin{tikzpicture}[baseline=(current bounding box.center)]
.....
\end{tikzpicture}
\end{equation}

```

$$\begin{array}{ccccc}
 A & \xrightarrow{a} & B & \xrightarrow{b} & C \\
 \downarrow c & & \downarrow d & & \downarrow e \\
 D & \xrightarrow{f} & E & \xrightarrow{g} & F
 \end{array} \tag{2}$$

After a while, tikz becomes addictive: The extended snake lemma says that the exact commutative diagram in blue gives rise to the exact sequence in red.

$$\begin{array}{ccccccc}
 0 & \longrightarrow & \text{Ker } f & \longrightarrow & \text{Ker } a & \longrightarrow & \text{Ker } b & \longrightarrow & \text{Ker } c & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \\
 & & A & \xrightarrow{f} & B & \xrightarrow{b} & C & \longrightarrow & 0 & & \\
 & & \downarrow a & & \downarrow d & & \downarrow c & & & & \\
 0 & \longrightarrow & A' & \xrightarrow{a} & B' & \xrightarrow{g'} & C' & \longrightarrow & 0 & & \\
 & & \downarrow & & \downarrow & & \downarrow & & & & \\
 & & \text{Coker } a & \longrightarrow & \text{Coker } b & \longrightarrow & \text{Coker } c & \longrightarrow & \text{Coker } g' & \longrightarrow & 0
 \end{array}$$

```

\begin{tikzpicture}[>=angle 90]
\matrix[matrix of math nodes,
row sep=3em, column sep=3em,,
text height=1.5ex, text depth=0.25ex]
{
|[name=00]| 0 & |[name=kf]| \text{Ker } f & |[name=ka]| \text{Ker } a & |[name=kb]| \text{Ker } b & |[name=kc]| \text{Ker } c & \\
& |[name=A]| A & |[name=B]| B & |[name=C]| C & |[name=01]| 0 & \\
& |[name=02]| 0 & |[name=A']| A' & |[name=B']| B' & |[name=C']| C' & \\
& |[name=ca]| \text{Coker } a & |[name=cb]| \text{Coker } b & |[name=cc]| \text{Coker } c & |[name=cg]| \text{Coker } g' & |[name=04]| 0
};
\draw[->,font=\scriptsize]
(ka) edge (A)
(kb) edge (B)
(kc) edge (C);
\draw[->,font=\scriptsize,blue,thick]
(A) edge node[auto] {$f$} (B)
(B) edge (C)
(C) edge (01)
(A) edge node[auto] {$a$} (A')
(B) edge node[auto] {$b$} (B')
(C) edge node[auto] {$c$} (C')

```

```

(02) edge (A')
(A') edge (B')
(B') edge node[auto] {$g'$} (C');
\draw[->,font=\scriptsize]
(A') edge (ca)
(B') edge (cb)
(C') edge (cc);
\draw[->,red]
(00) edge (kf)
(kf) edge (ka)
(ka) edge (kb)
(kb) edge (kc)
(ca) edge (cb)
(cb) edge (cc)
(cb) edge (cc)
(cc) edge (cg)
(cg) edge (04)
(kc) edge[out=0,in=180,red] node[above left] {$d$} (ca);
\end{tikzpicture}

```

For the last diagram, I added the following lines to the preamble

```

\usepackage{amsmath}
\DeclareMathOperator{\Coker}{Coker}
\DeclareMathOperator{\Ker}{Ker}

```